#This code will be used to demonstrate theorem 6.18. We will demonstrate that Q1= $\boldsymbol{\frac{x^2}{s^2}}$ ***is increasing along the $\gamma 2$ curve***; (***Part*** **1** $-$ ***locating potential critical points***)

#Define Q1;

$$Q1 := \frac{x^2 \cdot \left((z+w)^2 - y^2 \right)}{y^2 \cdot z \cdot w} :$$

#Apply the rational parametrization constructed in this chapter 4.

$$parametrization := \left\{ \mathrm{w} = \mathrm{a} + \frac{1}{a},\right.$$
$$z = b + \frac{1}{b},$$
$$x = a - \frac{1}{a} + b - \frac{1}{b},$$
$$\left. y = a - \frac{1}{a} - b + \frac{1}{b} \right\} :$$

$q1 := simplify(\ (\mathrm{expand}(\mathrm{subs}(\mathrm{parametrization}, \mathrm{Q1})))\ );$

$$q1 := \frac{4\,(b+a)^2\,(a\,b-1)^2}{(a^2+1)\,(b^2+1)\,(a-b)^2} \tag{1}$$

#Let us denote by D1=$\frac{da}{db}$*. By the chain rule, the derivative of Q1 is zero* if and *only* if

$$D1 := -\frac{diff\,(q1, b)}{diff\,(q1, a)} :$$

#Define r2

$$
\begin{aligned}
r2 := {} & \bigg(-1024 \left(b^2 - \frac{1}{2}\right) b^{13} \left(b^2 - 5\right) \left(b^2 + 1\right)^4 a^{23} + a^{32} b^{16} + \left(-4 b^{18} + 8 b^{16} - 4 b^{14}\right) a^2 + (\\
& -4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + \left(6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}\right) a^{28} + b^{16} - 9 \bigg(b^{16} \\
& + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 - 72 b^6 - 16 b^4 + 8 b^2 + 1 \bigg) b^8 a^{24} - 4 b^2 (b^{28} \\
& + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12} \\
& + 34120 b^{10} + 4315 b^8 + 1018 b^6 - 155 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} + 76 b^{26} - 539 b^{24} \\
& - 2118 b^{22} + 5211 b^{20} + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^8 \\
& - 518 b^6 + 37 b^4 + 12 b^2 + 1) a^{18} + 8 \bigg(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} \\
& + \frac{2163}{2} b^8 - 159 b^6 - 27 b^4 + b^2 + 1 \bigg) b^6 a^{22} - 1024 b^9 \left(b^2 + 1\right)^4 a^7 + 8 b^6 \bigg(b^{20} + b^{18} \\
& - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \bigg) \\
& a^{10} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \\
& - 27212 b^8 - 6620 b^6 - 2636 b^4 + 36 b^2 + 1) a^{12} + (-1024 b^{22} + 72 b^{20} + 692 b^{18} + 8 b^{16} \\
& - 76 b^{14} + 72 b^{12}) a^{26} + (-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} - 25752 b^{20} - 169584 b^{18} \\
& - 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^8 + 72 b^6 + 2 b^4) a^{20} + (b^{32} + 8 b^{30} \\
& + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16} \\
& - 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^8 - 120 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-9 b^{24} \\
& - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) a^8 \\
& + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\
& - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} \left(b^2 + 1\right)^4 a^{21} \\
& - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) \left(b^2 + 1\right)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6 \\
& + 81 b^4 - 22 b^2 + 1) \left(b^2 + 1\right)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\
& + 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 \left(b^2 + 1\right)^4 a^{13} + 2560 \bigg(b^6 - \frac{6}{5} b^4
\end{aligned}
$$

$$+\frac{8}{5}b^2-1\Big)b^7\left(b^2+1\right)^4a^{11}+512\,b^7\left(b^2-2\right)\left(b^2+1\right)^5a^9\Big):$$

#Using the implicit function theorem, compute $\frac{da}{db}=D2$ in *terms of* a and b

$$D2:=-\frac{diff(r2,b)}{diff(r2,a)}:$$

#We construct the curve $D2-D1$ *where its zeros describe points where the derivative of* $\frac{x^2}{s^2}$ *is zero*

$$f:=D2-D1:$$

#Calculate the numerator of f

$\tilde{f}:=\mathrm{numer}(f):$

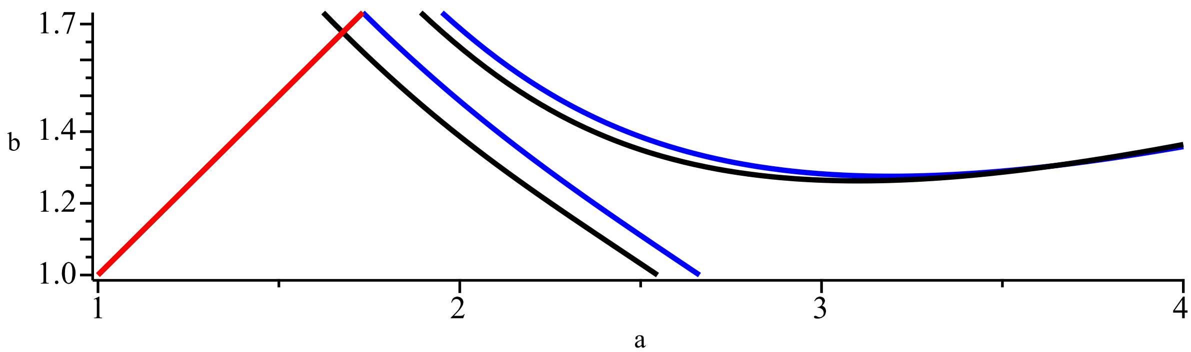
#Plot the curves r2=0 and $\tilde{f}$=0.

$r:=a-b:$

```
with(plots) :

implicitplot( [ r2 = 0, f̃ = 0, r = 0 ], a = 1 .. 4, b = 1 .. (sqrt(3) ),
            scaling = constrained,
            gridrefine = 2,
            thickness = 2,
            color = [blue, black, red],
            labels = ["a", "b"],
             title = "Graphic for r2=0 and  f̃ =0" );
```

Graphic for r2=0 and $\tilde{f}$=0



#Let's build the DescartesSingChanges function that counts the number of sign changes between the coefficients of a given polynomial.

```
DescartesSignChanges := proc(polynomial, variable)
  local coeffs, nonzero_coeffs, s, i, n;

  # Get the list of coefficients from highest degree to constant term
  coeffs := [seq(coeff(polynomial, variable, i), i = degree(polynomial, variable) .. 0, -1)];

  # Remove zero coefficients (they do not count in Descartes' Rule)
  nonzero_coeffs := remove(c -> c = 0, coeffs);

  # Count the number of sign changes
  n := nops(nonzero_coeffs);
  s := 0;
  for i from 1 to n-1 do
     if signum(nonzero_coeffs[i]) * signum(nonzero_coeffs[i + 1]) = -1 then
        s := s + 1;
     end if;
  end do;

  return s;
end proc:
```

#Compute the resultant between r2 and $\tilde{f}$ with respect to a

```
R1 := resultant( r2, f̃, a) :
```

#Compute the factors of the resultant R1 and count how many there are

```
dgb := proc(f)
  if type(f, `^`) then
    degree(op(1, f), b);
  else
    degree(f, b);
  end if;
end proc:

bfator := sort(
  [op(factor(R1))],
  (f, g) → dgb(f) < dgb(g)
 ) :

nops([op(factor(R1))]);
```

$$14 \tag{2}$$

#We list below the factors of the resultant R1, except for the factor 14 because it is too large. When necessary, we will use Sturm's algorithm to check if the factors cancel each other out within the interval [1, sqrt(3)) where γ2 is defined. For the rb14 factor, because of its high degree, we will use Moebius transformations as it requires less processing power.

```
rb1 := bfator[1];
```

$$rb1 := -1048868203031734503140222020224479475518024674359377 92 \tag{3}$$

```
rb2 := bfator[2];
```

$$rb2 := b^{672} \tag{4}$$

```
rb3 := bfator[3];
```

$$rb3 := \left(b^2 + 4\,b + 1\right)^2 \tag{5}$$

```
rb4 := bfator[4];
```

$$rb4 := \left(b^2 - 4\,b + 1\right)^2 \tag{6}$$

```
rb5 := bfator[5];
```

$$rb5 := \left(b^2 - 3\right)^3 \tag{7}$$

```
rb6 := bfator[6];
```

$$rb6 := \left(b^2 + 1\right)^{416} \tag{8}$$

```
rb7 := bfator[7];
sturm(sturmseq(rb7, b), b, 1, sqrt(3)); #number of roots in [1,sqrt(3))
```

$$rb7 := \left(b^4 - 4\,b^3 + 18\,b^2 - 4\,b + 1\right)^2$$

$$0 \quad (9)$$

rb8 := *bfator*[8];

$$rb8 := \left(b^4 + 4\,b^3 + 18\,b^2 + 4\,b + 1\right)^2 \quad (10)$$

rb9 := *bfator*[9];
sturm(*sturmseq*(*rb9*, *b*), *b*, 1, sqrt(3)); *#number of roots in [1,sqrt*(3))

$$rb9 := \left(7\,b^4 - 6\,b^2 + 3\right)^3$$

$$0 \quad (11)$$

rb10 := *bfator*[10];
sturm(*sturmseq*(*rb10*, *b*), *b*, 1, sqrt(3)); *#number of roots in [1,sqrt*(3))

$$rb10 := 15\,b^6 - 3\,b^4 + 45\,b^2 - 1$$

$$0 \quad (12)$$

rb11 := *bfator*[11];

$$rb11 := 17\,b^6 + 3\,b^4 + 51\,b^2 + 1 \quad (13)$$

rb12 := *bfator*[12];
sturm(*sturmseq*(*rb12*, *b*), *b*, 1, sqrt(3)); *#number of roots in [1,sqrt*(3))

$$rb12 := \left(47\,b^6 - 195\,b^4 + 141\,b^2 - 65\right)^3$$

$$0 \quad (14)$$

rb13 := *bfator*[13];
sturm(*sturmseq*(*op*(1, *rb13*), *b*), *b*, 1, (sqrt(3))); *#number of roots in [1,sqrt*(3))

$$rb13 := \left(45241\,b^{48} - 983136\,b^{46} + 8765868\,b^{44} - 53430800\,b^{42} + 292102794\,b^{40} - 1327146624\,b^{38} + 4932365356\,b^{36} - 14813913168\,b^{34} + 36572797215\,b^{32} - 74566977600\,b^{30} + 126486144024\,b^{28} - 177535429920\,b^{26} + 204746625068\,b^{24} - 190367984256\,b^{22} + 140202896376\,b^{20} - 79729549344\,b^{18} + 34666181511\,b^{16} - 11137697376\,b^{14} + 2222925500\,b^{12} - 25011408\,b^{10} - 137706774\,b^{8} + 38273024\,b^{6} - 2821284\,b^{4} - 1082256\,b^{2} + 374017\right)^2$$

$$0 \quad (15)$$

#The factor rb14 has degree 376, so we will not show it, just compute how many zeros it has in the desired range.

rb14 := *bfator*[14]:

degree(rb14, *b*);

$$376 \quad (16)$$

#From previous tests, using the Moebius transformation on the rb14 factor was more efficient when subdividing the interval [1, sqrt(3)) into two subintervals: [1, 1.4) **and** [1.14, sqrt(3)).

#Apply the Möbius transformation on the interval (1, 1.4), constructing Rb14_1

$$bmoebius1 := \left\{ b = \frac{1 + \frac{7}{5} \cdot d}{1 + d} \right\} :$$

$Rb14\_1 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{bmoebius1}, \text{rb14}))), d) :$

#From here we will analyze the signal variation of all coefficients Rb14_1

$DescartesSignChanges(Rb14\_1, d);$

$$2 \tag{17}$$

#Apply the Möbius transformation on the interval [1.4, sqrt(3)), constructing Rb14_2

$$bmoebius2 := \left\{ b = \frac{\frac{7}{5} + \text{sqrt}(3) \cdot \text{d}}{1 + d} \right\} :$$

$Rb14\_2 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{bmoebius2}, \text{rb14}))), d) :$

#From here we will analyze the signal variation of all coefficients Rb14_2

$DescartesSignChanges(Rb14\_2, d);$

$$0 \tag{18}$$

#Finding possible candidates for zeros of the factor rb14

$fsolve(rb14 = 0, b, 1 .. (\text{sqrt}(3)));$

$$1.066555212, 1.304932132 \tag{19}$$

#We now repeat the same process for the resultant of r2 with f̃, in *relation* **to** *b*

$R2 := resultant(\mathrm{r2}, \tilde{\mathrm{f}}, \mathrm{b}):$

#Compute the factors of the resultant R2 and count how many there are

```
dg(a) := proc(f)
  if type(f, `^`) then
    degree(op(1, f), a);
  else
    degree(f, a);
  end if;
end proc:

afator := sort(
  [op(factor(R2))],
  (f, g) -> dg(a)(f) < dg(a)(g)
):
```

$nops([op(factor(R2))]);$

$$14 \tag{20}$$

#We list below the factors of the resultant R2, except for the factor 14 because it is too large. When necessary, we will use Sturm's algorithm to check if the factors cancel each other out within the interval [sqrt(3), 4) where γ2 is defined. For the ra14 factor, because of its high degree, we will use Moebius transformations as it requires less processing power.

$ra1 := afator[1];$

$$ra1 := -10488682030317345031402220202244794755180246743593779\!2 \tag{21}$$

$ra2 := afator[2];$

$$ra2 := a^{672} \tag{22}$$

$ra3 := afator[3];$

$$ra3 := (a^2 + 4a + 1)^2 \tag{23}$$

$ra4 := afator[4];$

$$ra4 := (a^2 - 4a + 1)^2 \tag{24}$$

$ra5 := afator[5];$

$$ra5 := (a^2 - 3)^3 \tag{25}$$

$ra6 := afator[6];$

$$ra6 := (a^2 + 1)^{416} \tag{26}$$

$ra7 := afator[7];$
$sturm(sturmseq(ra7, a), a, \mathrm{sqrt}(3), 4);$ *#number of roots in [sqrt(3), 4)*

$$ra7 := \left(a^4 - 4a^3 + 18a^2 - 4a + 1\right)^2$$
$$0 \tag{27}$$

$ra8 := afator[8];$

$$ra8 := \left(a^4 + 4a^3 + 18a^2 + 4a + 1\right)^2 \tag{28}$$

$ra9 := afator[9];$
$sturm(sturmseq(ra9, a), a, \mathrm{sqrt}(3), 4);$ *#number of roots in [sqrt(3), 4)*

$$ra9 := \left(7a^4 - 6a^2 + 3\right)^3$$
$$0 \tag{29}$$

$ra10 := afator[10];$

$$ra10 := 15a^6 - 3a^4 + 45a^2 - 1 \tag{30}$$

$ra11 := afator[11];$

$$ra11 := \left(a^4 + 4a^3 + 18a^2 + 4a + 1\right)^2 \tag{31}$$

$ra12 := afator[12];$
$sturm(sturmseq(ra12, a), a, \mathrm{sqrt}(3), 4);$ *#number of roots in [sqrt(3), 4)*

$$ra12 := \left(47a^6 - 195a^4 + 141a^2 - 65\right)^3$$
$$1 \tag{32}$$

#Finding possible candidates for zeros of the factor ra12

$fsolve(op(1, ra12) = 0, a, \mathrm{sqrt}(3) .. 4);$

$$1.839289820 \tag{33}$$

$ra13 := afator[13];$
$sturm(sturmseq(ra13, a), a, \mathrm{sqrt}(3), 4);$ *#number of roots in [sqrt(3), 4)*

$$\begin{aligned} ra13 := \big(&175559\,a^{48} - 1769112\,a^{46} + 6749652\,a^{44} + 9460584\,a^{42} - 193871370\,a^{40} \\ &+ 750687384\,a^{38} - 111548076\,a^{36} - 9218817960\,a^{34} + 37541503329\,a^{32} \\ &- 86577381616\,a^{30} + 144267501288\,a^{28} - 188241754608\,a^{26} + 198904792788\,a^{24} \\ &- 173518719696\,a^{22} + 126058415880\,a^{20} - 76187388496\,a^{18} + 37964657145\,a^{16} \\ &- 15394822392\,a^{14} + 5027018436\,a^{12} - 1306370040\,a^{10} + 287741334\,a^{8} - 59068744\,a^{6} \\ &+ 10970148\,a^{4} - 1480584\,a^{2} + 96383\big)^2 \end{aligned}$$
$$0 \tag{34}$$

#The factor ra14 has degree 376, so we will not show it, just compute how many zeros it has in the desired range.

$ra14 := afator[14]:$

$degree(\mathrm{ra14}, \mathrm{a});$

$$376 \tag{35}$$

#Apply the Möbius transformation on the interval (sqrt(3), 4$\Big)$, constructing Ra14

$$amoebius := \left\{a = \frac{\text{sqrt}(3) + 4 \cdot c}{1 + c}\right\}:$$

Ra14 := collect(numer(expand(subs(amoebius, ra14))), c) :

#From here we will analyze the signal variation of all coefficients Ra14

DescartesSignChanges(*Ra14*, *c*);

376

5 **(36)**

#Finding possible candidates for zeros of the factor ra13

fsolve(*ra14* = 0, *a*, sqrt(3) .. 4);

3.192177604, 3.538902695, 3.555974116, 3.643966007, 3.884436541 **(37)**

#Let's rule out that the root a=2+sqrt(3) does not form a pair with any of the other candidates b

#Apply the Möbius transformation on the interval [1, 2+sqrt(3)), constructing $\hat{r}1$

$$moebius := \left\{ b = \frac{1 + \mathrm{sqrt}(3)\, d}{1 + d} \right\}:$$

$\hat{r}2 := \mathrm{collect}(\mathrm{numer}(\mathrm{expand}(\mathrm{subs}(\mathrm{moebius}, eval(r2, a = 2 + \mathrm{sqrt}(3))))), d):$

#From here we will analyze the signal variation of all coefficients $\hat{r}1$

$DescartesSignChanges(\hat{r}2, \mathrm{d});$

$$1 \quad \textbf{(38)}$$

$fsolve(eval(r2, a = 2 + \mathrm{sqrt}(3)) = 0, b, 1 .. \mathrm{sqrt}(3))$

$$1.316165044 \quad \textbf{(39)}$$